MILITARY SPECIFICATION

FREQUENCY CHANGER, SOLID STATE, AIR COOLED (NAVAL SHIPBOARD)

This specification is approved for use by the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers solid state, air cooled frequency changers for the conversion of electrical power from 60 hertz (Hz), 3-phase, to 400 Hz, 3-phase, for non-nuclear power naval surface ship use.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards and handbook. Unless otherwise specified, the following specifications, standards and handbook of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

PPP-P-40 - Packaging and Packing of Hand Tools.

MILITARY

MIL-B-117 - Bags, Sleeves and Tubing - Interior Packaging.
MIL-S-901 - Shock Tests, H.I. (High-Impact); Shipboard Machinery, Equipment and Systems, Requirements for.
MIL-C-915 - Cable and Cord Electrical, for Shipboard Use General Specification for.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Sea Systems Command, SEA 55Z3, Department of the Navy, Washington, DC 20362 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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MIL-C-915/32 - Cable, Electrical, 1000 Volts, Type MSCU (Including Variations MSCA and MSCS).
MIL-E-917 - Electric Power Equipment, Basic Requirements (Naval Shipboard Use).
MIL-E-2036 - Enclosures for Electric and Electronic Equipment, Naval Shipboard.
MIL-M-7793 - Meter, Time Totalizing.
MIL-P-15024 - Plates, Tags and Bands for Identification of Equipment.
MIL-P-15024/5 - Plates, Identification.
MIL-F-15160 - Fuses: Instrument, Power and Telephone.
MIL-M-16034 - Meters, Electrical-Indicating (Switchboard and Portable Types).
MIL-M-16034/9 - Voltmeters, A.C. and D.C., Portable, 0.25-Percent Accuracy Class.
MIL-E-17555 - Electronic and Electrical Equipment, Accessories, and Repair Parts; Packaging and Packing of.
MIL-S-19500 - Semiconductor Devices, General Specification for.
MIL-M-38510 - Microcircuits, General Specification for.

STANDARDS

MILITARY

MIL-STD-108 - Definitions of and Basic Requirements for Enclosures for Electric and Electronic Equipment.
MIL-STD-167-1 - Mechanical Vibrations of Shipboard Equipment (Type I - Environmental and Type II - Internally Excited).
MIL-STD-202 - Test Methods for Electronic and Electrical Component Parts.
MIL-STD-454 - Standard General Requirements for Electronic Equipment.
MIL-STD-461 - Electromagnetic Emission and Susceptibility Requirements for the Control of Electromagnetic Interference.
MIL-STD-462 - Electromagnetic Interference Characteristics, Measurement of.
MIL-STD-471 - Maintainability Verification/Demonstration/Evaluation.
MIL-STD-740 - Airborne and Structureborne Noise Measurements and Acceptance Criteria of Shipboard Equipment.
MIL-STD-781 - Reliability Tests, Exponential Distribution.
MIL-STD-810 - Environmental Test Methods and Engineering Guidelines.
MIL-STD-883 - Test Methods and Procedures for Microelectronics.
MIL-STD-965 - Parts Control Program.
MIL-STD-1310 - Shipboard Bonding, Grounding, and Other Techniques for Electromagnetic Compatibility and Safety.
DOD-STD-1399, Section 300 - Interface Standard for Shipboard Systems, Electric Power, Alternating Current, (Metric).

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MIL-STD-1472 - Human Engineering Design Criteria for Military Systems, Equipment and Facilities.
MIL-STD-1629 - Procedures for Performing a Failure Mode, Effects and Criticality Analysis.
MIL-STD-2068 - Reliability Development Tests.

HANDBOOK

MILITARY

MIL-HDBK-217 - Reliability Prediction of Electronic Equipment.

2.1.2 Government drawings and publications. The following Government drawing and publications form a part of this specification to the extent specified herein.

DRAWING

NAVAL SEA SYSTEMS COMMAND (NAVSEA)

NAVSEA 46D1931-2 - Top Assembly 60/400 Hz Air Cooled Power Converter.

PUBLICATIONS

NAVAL SEA SYSTEMS COMMAND (NAVSEA)

0967-LP-312-8010 - Electronic Equipment, Shipboard, Maintainability, Design for Criteria Handbook for Designers of.
0967-LP-597-1011 - Electronic Equipment, Parts Application and Reliability Information for Navy.

(Copies of specifications, standards, handbooks, drawings and publications required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following document forms a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

INSTITUTE OF ELECTRICAL AND ELECTRONICS ENGINEERS (IEEE)

IEEE 200 - Reference Designations for Electrical and Electronics Parts and Equipment. (DoD adopted)

(Application for copies should be addressed to the Institute of Electrical and Electronics Engineers, 345 East 47th Street, New York, NY 10017.)

(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

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2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence; with the exception of the NAVSEA drawing package defined by NAVSEA 46D1931-2. If conflict exists between NAVSEA Drawing 46D1931-2 and this specification, NAVSEA should be consulted.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2.1), a sample shall be subjected to first article inspection (see 4.4 and 6.3).

3.2 Performance.

3.2.1 Input power. The input to the equipment (see 6.4.1) shall be 440 volts (V), 60 Hz, 3-phase, 3-wire, ungrounded, type I power complying with the characteristics and ranges as specified in DOD-STD-1399, section 300. Equipment shall not be damaged when the input voltage phase rotation is in a reverse order (AC-CB-BA).

3.2.1.1 Self-protection from input voltage variation and transient. When the equipment is supplying any load from no-load to rated-load, the equipment shall withstand, without damage or blowing of fuses, any input voltage or frequency variation or both outside the transient limits specified in DOD-STD-1399, section 300. When the input voltage and frequency exceed the input transient limits, they may take any of the following conditions:

(a) The input voltage instantly drops to zero.
(b) The input voltage drops between 0 and 82 percent of nominal and input frequency drops between 0 and 92 percent of nominal. The voltage-frequency relation during the period shall be such that voltage will be maintained at nominal until the frequency drops down to about 83 percent of nominal, then the voltage drops in proportion to the frequency at a frequency less than 83 percent, and finally the input voltage collapses to zero at a frequency about 35 percent.
(c) The input voltage rises to 150 percent of nominal or input frequency rises to 125 percent of nominal, or both.

Upon return of the input voltage and frequency to the specified range (see 3.2.1), the equipment shall operate and supply the load with the same output voltage and frequency as that prior to the input power changes. The performance of the equipment shall meet all requirements of this specification without any adjustment or replacement of parts. If the equipment has shut off, normal starting procedure (see 3.2.1.5) shall be used to restart the equipment.

3.2.1.2 Self-protection from input spikes. When the equipment input voltage experiences a transient (line-to-line or line-to-ground, or both) up to 2500 V, having a basic impulse level (BIL) wave shape as shown on figure 1, the equipment output voltage waveform shall not deviate from that specified in 3.2.2.8. Equipment shall continue to operate without shutting down or malfunction. The calculated 60 Hz source surge impedance shall be as specified in table I.

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TABLE I. Calculated 60 Hz source surge impedance.

Navy type cable	Current rating of cable in amperes at 50°C	Surge impedance in ohms
T14	47	26
T23	64	26
T50	101	20
T75	136	20
T100	160	19
T150	216	18
T200	250	18
T300	320	18
T400	400	17

3.2.1.3 Input current harmonics. When the equipment is powered by a 60 Hz sinusoidal voltage of the specified range (see 3.2.1) and with the equipment supplying any load from no-load to rated-load, the total current harmonic content shall be not more than 5 percent of the rated-load input current. Individual harmonic currents from the 2nd harmonic through the 11th harmonic shall be not more than 4.5 percent. Individual harmonics above the 11th harmonic until the 333 harmonic, shall be not more than 49.5/n percent, where n is the number of the harmonic.

3.2.1.4 Input current balance. When the equipment is supplying any load from 10 percent to rated-load and input 3-phase voltages are balanced within 4 V, the input current shall be balanced to the extent that current in any input line shall not differ by more than 5 percent of the rated-load input current from the arithmetic average current of the three input lines. When the equipment is supplying any load from 10 percent to rated-load and input 3-phase voltages are unbalanced by 3 percent as specified in DOD-STD-1399, section 300, the input current shall be balanced to the extent that the current in any input line shall not differ by more than 10 percent of the rated-load input current from the arithmetic average current of the three input lines. The 60 Hz source impedance shall be as specified in table II. The 60 Hz source impedance for the 110 kilowatt (kW), and the 185 kW units shall be as specified in table II.

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TABLE II. 60 Hz, maximum power source impedance in ohms.1/

Load power factor	1.0		0.8		0.7		0.6		0.5		0.4	
Load kVA 3-phase	R	X	R	X	R	X	R	X	R	X	R	X
440-V, 3-phase, 60 Hz												
10	0.81	0.058	0.97	0.058	1.1	0.058	1.2	0.077	1.4	0.077	1.5	0.077
20	.42	.058	.51	.058	0.57	.058	0.64	.067	0.70	.067	0.81	.067
30	.30	.058	.36	.058	.39	.058	.44	.058	.49	.071	.54	.071
40	.23	.058	.28	.058	.29	.058	.33	.058	.37	.067	.42	.067
50	.19	.058	.23	.058	.23	.058	.27	.058	.29	.058	.34	.069
60	.17	.061	.19	.061	.22	.061	.22	.061	.25	.061	.28	.071
70	.15	.060	.17	.060	.18	.069	.20	.069	.22	.069	.24	.069
80	.14	.060	.15	.070	.16	.070	.17	.070	.19	.070	.21	.070
90	.14	.060	.14	.060	.15	.060	.16	.069	.17	.069	.19	.069
100	.12	.060	.13	.060	.14	.069	.15	.069	.16	.069	.17	.069

1/ To convert to per unit values:

$$X_{pu} = \frac{X(\text{ohms})\ (\text{kVA of load})\ 1000}{(\text{L-L voltage})^2}$$

3.2.1.5 Starting time. Equipment shall accept any load from no-load to rated-load and operates in compliance with requirements specified herein, in not more than 5 seconds after start. Equipment shall not require adjustments to meet these requirements, when the input voltage and frequency varies within the range specified in 3.2.1.

3.2.1.5.1 Prohibition of start. Equipment shall start when the input voltage is within the range of 409 V to 471 V and the input frequency is within the range of 57 to 63 Hz. Equipment shall not be started when the input voltage, or the input frequency is outside the range where output power can be maintained (see 3.2.7.5).

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3.2.1.6 Starting input current. Input current shall be limited during the starting time (see 3.2.1.5). The peak input starting (or in-rush) current shall be not greater than 150 percent of the peak full-load steady state input current.

3.2.1.7 No-load input current. The maximum no-load input current shall be not greater than 10 percent of rated-load input current.

3.2.1.8 Grounding. Equipment shall operate as specified herein, when connected to an ungrounded ship's input power system having 3 to 20 microfarad (μF) capacitance from any line to ground, or with any single input or output line or terminal intermittently or permanently grounded.

3.2.1.9 Input power factor. The input power factor presented to the 60 Hz input power by the equipment, shall be between 0.75 lagging and 0.9 leading.

3.2.2 Output voltage. Rated output voltage shall be 450 volts root mean square (Vrms). The output circuit shall be delta-connected, ungrounded, 3-phase, 3-wire, 400 Hz for all units. The phase rotation of the output voltage shall be a clockwise sequence of AB-BC-CA.

3.2.2.1 Voltage regulation. The average of the volt root mean square (Vrms) values of the output voltages shall be maintained at the regulated point (the equipment output terminals (see 3.2.2.2)) within plus or minus 0.5 percent of rated output voltage, under the following variable conditions and any combination thereof, at all voltage settings within the adjustment range as specified in 3.2.2.3:

(a) A voltage drop from the terminals of the equipment to the regulated point of any amount from 0 to 7 percent of rated output voltage, as specified in 3.2.2.2.
(b) A load variation from no-load to 125 percent of rated-load.
(c) Load power factor varying between 0.6 lagging and 0.9 leading.
(d) An input voltage and frequency variation within the range specified herein.
(e) Variation of the equipment temperature from cold start (at ambient temperature ranging from 0 to 50 degrees Celsius (°C) to maximum operating temperature (at ambient temperature ranging from 0 to 50°C).
(f) Under the unbalanced load conditions specified in 3.2.2.11.

3.2.2.2 Line drop compensation. The converter shall be capable of compensating for a full-load line voltage drop to the load of 7 percent. While operating in the line drop compensation mode, the converter shall meet the voltage regulation requirements of 3.2.2.1 at the load terminals. All other performance requirements of this specification shall be met at the converter output terminals. Line drop compensation shall not be applied to converters operating in parallel.

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3.2.2.3 Voltage adjustment. Means shall be provided to adjust the output voltage at the regulated point over a range of plus or minus 5 percent of the rated voltage, with adjustment resolution not more than 50 millivolts at any load from no-load to 125 percent rated-load under any steady state input conditions. The adjustment device shall have a slotted shaft for screw-driver type adjustment. A voltmeter shall be provided which will indicate the unit output voltages.

3.2.2.4 Transient output voltage and recovery. The sudden change of 0.9 lagging power factor loads on the equipment from no-load to half-load, from half-load to rated-load, from rated-load to half-load, and from three-fourths load to one-fourth load shall cause an output voltage deviation of not more than plus or minus 5.0 percent of rated voltage. The voltage at the equipment output terminals shall recover and stay within the steady state regulation band (see 3.2.2.1), within 0.025 second. Circuits shall be so designed that voltage transients developed within the equipment shall not reflect back into the input power system such that transient limits as specified in DOD-STD-1399, section 300, are exceeded for alternating current (a.c.) input (type I power).

3.2.2.5 Frequency regulation. Output frequency shall be maintained within plus or minus 0.1 percent of rated frequency under the following conditions or combination thereof:

(a) A load variation from no-load to rated-load.
(b) Load power factor varying between 0.6 lagging and 0.9 leading.
(c) An input voltage and frequency variation within the range specified herein.
(d) Variation of the equipment temperature from cold start (at ambient temperature ranging from 0 to 50°C.) to maximum operating temperature (at ambient temperature ranging from 0 to 50°C).
(e) Variation of output voltage specified in 3.2.2.3.

3.2.2.6 Transient frequency deviation and recovery. A sudden change of 0.9 lagging power factor loads on the equipment from no-load to rated-load and from rated-load to no-load, shall not cause a frequency deviation greater than the steady state band specified in 3.2.2.5.

3.2.2.7 Amplitude and frequency modulation. Amplitude and frequency modulation of the output voltage (as specified in DOD-STD-1399, section 300) shall not exceed 1 percent and 0.5 percent, respectively, under any of the conditions or combinations of conditions as specified in 3.2.2.5.

3.2.2.8 Output voltage waveform. Under any load conditions, between no-load to rated-load, the root mean square (rms) value of all of the harmonics in the output voltage shall not exceed 3 percent of the rated output voltage. The rms value of any single harmonic shall not exceed 2 percent of the rms value of the rated output voltage. The deviation factor (as specified in DOD-STD-1399, section 300) of the output voltage waveform shall not exceed 5 percent. Peak-to-peak amplitude of voltage spikes (between the phases) of any frequencies shall not exceed 3 percent of the rms value of the rated output voltage.

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3.2.2.9 Output phase displacement. Displacement angle between adjacent voltage phases shall be 120 ± 1 degrees under all conditions of operation specified herein.

3.2.2.10 Output voltage balance under balanced load conditions. The difference between phase voltages at the voltage regulated point of the equipment shall be not greater than 0.5 percent of rated voltage at no-load and at any balanced 3-phase load up to rated current with any power factor between 0.6 lagging and 0.9 leading.

3.2.2.11 Output voltage balance under unbalanced load conditions. The voltage variation between phases at the equipment output terminals shall be not greater than 1 percent of the rated voltage under the unbalanced load conditions as shown in table III, when the output voltage is regulated at the equipment output terminals.

TABLE III. Unbalanced loads.

Load conditions	Line A (percent of rated current)	Line B (percent of rated current)	Line C (percent of rated current)	Power factor
1	90	75	75	0.6 lagging to unity
2	75	90	75	0.6 lagging to unity
3	75	75	90	0.6 lagging to unity

Equipment shall operate with a single phase 0.6 power factor load carrying a current 15 percent of rated current. The voltage of the loaded phase at the voltage regulated point shall have regulation as specified in 3.2.2.1.

3.2.2.12 Input voltage change effect. When the equipment is supplying any load from no-load to rated-load and with any output and input steady state voltage and frequency within the range specified in 3.2.1, the occurrence of any input transient voltage up to plus or minus 20 percent of nominal, with a recovery time of 2 seconds to the set limits, as specified for type I power in DOD-STD-1399, section 300, shall not cause an output voltage deviation of plus or minus 3 percent of the initial value. The output voltage shall recover to and remain within the steady state regulation band, within 0.02 seconds from the termination of the input voltage change.

3.2.3 Power ratings. The power rating of the equipment shall be 110 kW at 0.9 lagging power factor. The kilovoltampere (kVA) rating and power factor rating shall be within the power range (shaded area) as shown on figure 2.

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3.2.3.1 Duty. Equipment shall operate continuously at the specified power rating within the power factor range as shown on figure 2.

3.2.4 Efficiency. The overall efficiency of the equipment shall be a minimum of 86 percent when operating at rated input voltage, frequency, load and power factor.

3.2.5 No-load losses. No-load losses of the equipment shall be not more than 60 percent of the rated-load losses at nominal input voltage and frequency.

3.2.6 Overload. Equipment shall provide and sustain an overload current of 125 percent of rated current at the rated output voltage for a minimum of 30 minutes and meet all requirements specified herein. The output voltage regulation for any overload current up to 125 percent of the rated output voltage shall be as specified in 3.2.2.1. The equipment shall automatically shut down if critical component or air temperatures are exceeded. An indicator shall be provided to show temperature shut down.

3.2.7 Fault and malfunction protection.

3.2.7.1 Fault isolation. Equipment shall incorporate fault isolation for each module or drawer (see 6.4.4 and 6.4.5), to prevent cascading to other modules or drawers. A shorted or noncommutated power semiconductor shall not result in damage to any other part (see 6.4.6) of the equipment, except that fuses may blow.

3.2.7.2 Fault current. Output current between 125 and 150 percent of rated current is considered fault current. Equipment shall provide and sustain any fault current up to 150 percent of rated current for a minimum of 15 minutes when a fault develops at any external circuits supplied by the equipment.

3.2.7.3 Short circuit protection. Output current above 150 percent of rated current shall be considered short circuit current. Equipment shall protect itself without damage to any parts from any short circuit current on any or all output phases. Protection shall be provided by current limiting to a value not less than 150 percent of rated current. The equipment shall deliver at least 275 percent of rated-load current into a short circuit, and when the short circuit current exceeds 275 percent of rated-load current, the equipment shall automatically shut down after a time interval of not less than 1.5 seconds. An indicator shall be provided to show that a short circuit occurred. On failure of the current limiting circuits, the equipment shall be protected by fuses (see 3.2.7.4).

3.2.7.4 Internal circuit protection. Fuses shall be provided for power circuits and semiconductors as backup for the current limiting protection (see 3.2.7.3). Each drawer containing power switching semiconductors shall be separately fused. Circuits with filter capacitors shall have fuse protection against capacitor failure.

3.2.7.5 Protection for utilization equipment. A protective scheme shall be provided to shut down the equipment if the output voltage shown in table IV exists. The equipment shut-down shall be delayed from a minimum of 45 milliseconds (ms) to a maximum of 250 ms to prevent nuisance tripping because of

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possible system transients. An alarm light shall indicate the overvoltage condition. In addition, an alarm light shall indicate an output undervoltage condition of 425 V or less. Both alarm lights shall latch until manually reset.

TABLE IV. Shut-down voltage setting.

Condition	Setting (percent of rated output)	Tolerance (percent of rated output)
Over-voltage	110	Plus 4, minus 0

3.2.8 Parallel operation. Equipment shall provide paralleling operation capability at the levels of reliability and maintainability specified herein. It shall be possible to parallel up to a maximum of six units, without interrupting the load and without introducing transient voltages at the input line and output load bus outside the specified limits. After the units are connected for parallel operation and the output voltage of each unit has been manually adjusted (see 3.2.2.3) so that the maximum terminal voltage difference between the units is not more than 1 V, the synchronization and parallel operation of the units shall be automatically accomplished to meet all requirements specified herein. The units shall parallel under any load conditions from no-load to rated-load.

3.2.8.1 Synchronization. After the units are connected for parallel operation, synchronization shall take place automatically, with no time lag after connection.

3.2.8.2 Real and reactive load division. When paralleling identical size units, the maximum difference in kW supplied by the units in parallel shall not exceed 10 percent of the kW rating of one of the units under load conditions between 10 and 100 percent of the total rated-load of the paralleled units. The maximum difference in kilovar (kvar) supplied by the units operating in parallel shall not exceed 10 percent of the kvar rating of one of the units under any load condition between 10 and 100 percent of the total rated-load of the paralleled units. When paralleling different size units, the output of each unit shall be directly proportioned to the kW rating of the individual unit. The maximum difference in kW between the actual kW supplied by each unit and its proportional share of the total kW, shall not exceed plus or minus 5 percent of the average kW rating of units in parallel under load conditions between 10 and 100 percent of the total rated-load of the paralleled units. The maximum difference in kvar between the actual kvar supplied by each unit and its proportional share of the total kvar shall not exceed plus or minus 5 percent of the average kvar rating of units in parallel under load conditions between 10 and 100 percent of the total rated-load of the paralleled units. The above requirements shall apply for input voltage variations specified in 3.2.1 and other environmental requirements specified herein.

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3.2.8.3 Selective tripping. Equipment shall operate so that when any malfunction occurs to any one of the units operated in parallel, the malfunctioned unit shall automatically shut itself down and the remaining units shall continue to operate in parallel, as specified herein.

3.2.8.4 Arrangement of parallel operation. The parallel operation of up to the maximum six units shall be connected to a common bus (see 3.2.8.4.1) or to a split bus (see 3.2.8.4.2) arrangement. Paralleling requirements as specified in 3.2.8, 3.2.8.1, 3.2.8.2 and 3.2.8.3 shall apply to both the common bus and the split bus arrangements.

3.2.8.4.1 Common bus parallel operation. The common bus parallel operation shall be connected as shown on figure 3 (figure 3 only shows two units). When the 400 Hz bus is located in the same compartment with the frequency changers such that the impedance of the 400 Hz cable between the units and the 400 Hz bus shall be considered negligible, the voltage of the common bus shall meet all output requirements specified herein (see 3.2.2 through 3.2.2.12).

3.2.8.4.2 Split bus parallel operation. The split bus parallel operation shall be connected as shown on figure 4 (figure 4 only shows two units and two buses) up to the maximum of six units and three buses in any combination. When each unit is located close to its own 400 Hz bus such that the impedance of the 400 Hz cable between the unit and its bus shall be considered negligible and the maximum impedance of each bus tie cable is less than 2 percent (based on the rated output current of one unit), the voltage at one of the 400 Hz buses as A, B or C (bus C is not shown on figure 4) shall meet all output requirements specified herein (see 3.2.2 through 3.2.2.12). Selection of the bus to meet the output voltage requirements may be accomplished by a control cable connecting between the unit and its bus. One normally-open and one normally-closed auxiliary contact are available in each bus tie breaker and can be used to provide bus tie breaker status information, for the control of the split bus parallel operation.

3.3 Design and construction.

3.3.1 General requirements. In addition to the requirements of this specification and NAVSEA Drawing 46D1931-1, the equipment shall be in accordance with MIL-E-917 except the following:

(a) Fundamental equipment characteristics (excluding safety)
(b) Grounding
(c) Input power variations (a.c. equipment)
(d) Inclined operation
(e) Ambient temperature
(f) Impact shock and vibration
(g) Equipment mounting
(h) Radio interference
(i) Noise reduction for submarine
(j) Piping system
(k) Drawings
(l) Manuals
(m) Workmanship

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Whenever a requirement in accordance with MIL-E-917 conflicts with a requirement of this specification, the requirement of this specification shall govern.

3.3.2 Electrical designs. Equipment shall utilize building block module concept with solid state circuitry and devices only. Electro-mechanical type relays may be used only under the conditions applicable to the use of nonstandard parts (see 3.4.1). Vacuum tubes or other similar thermonic devices shall not be used. Control and logic circuits shall have a common negative. Control wires shall be separated from the power conductors to avoid noise pickup in the control circuits. Electrical isolation between control circuits and power semiconductor devices by optical coupling may be considered.

3.3.2.1 Isolation. Equipment input shall be electrically isolated from the equipment output. Isolation transformers shall have electrically isolated windings (primary and secondary windings). Auto-transformers shall not be used.

3.3.2.2 Input and output transformers. Equipment input and output transformers shall be a balanced arrangement (for example: 3-phase Δ - connection, 3-phase Y connection, 6-phase star connection, and so forth). Scott "T" and open-delta connection shall not be used.

3.3.2.3 Dielectric withstanding voltage and insulation resistance. Equipment, except the control drawer, shall pass insulation resistance and dielectric withstanding voltage tests as specified in 4.7.3 and 4.7.4. Insulation resistance of the circuits shall be not less than 10 megohms when measured with units at ambient temperature of 25 plus or minus 2°C. Dielectric strength shall be twice rated plus 1000 V for 60 seconds. The design of the equipment shall be such that required insulation resistance and dielectric withstanding voltage tests on the insulation system can be conducted without either incipient or catastrophic damage or without significant (within acceptable product variability tolerance) degradation to any material or part, including the electrical insulation. Futhermore, a defect in the insulation system shall not cause either of the tests, when properly conducted, to result in damage or degradation to any other material or part except that which possibly may be inflicted to the immediately adjacent parts separated by the defective insulation.

3.3.3 Size and weight. The equipment including air plenums, shall not exceed the weight and dimensions shown in table V.

TABLE V. Weights and dimensions.

Power rating	Dimensions			Maximum weight per unit (pounds)
	Width (inches)	Depth (inches)	Height (inches)	
110 kW	72	38	76	5,900

Weight of equipment shall be kept to a minimum. At delivery, the net weight and also the operating weight with fluids, where applicable, shall be indicated. In addition, the vertical, longitudinal, and transverse location of the center of gravity (wet or dry, as appropriate) of each equipment weighing 50 kilograms (100 pounds) or more, shall be shown on the outline or assembly drawing.

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3.3.4 Cabinet enclosure. Equipment shall have a dripproof protected (45-degree) enclosure as specified in MIL-STD-108.

3.3.5 Ground connection. Exposed metal parts of the cabinets shall be at ground (ship's hull) potential at all times. Exposed metal portion of electrical parts (such as switches and rheostats) which is not part of the electrical circuits, shall be electrically grounded to the frame of the cabinet. A ground stud shall be provided on each cabinet as specified in MIL-STD-1310 to accept a class C, type II bond strap which will be supplied and installed by the contracting agency. The ground stud shall be clearly marked and shall be readily accessible and so located or protected as to reduce the probability of breakage during transit or operation. Leakage current through filter capacitors to ground shall be limited to 30 milliamperes per phase.

3.3.6 Cabinet mounted control. An elapsed time meter and the output voltage adjustment device (see 3.2.2.3) shall be mounted on the cabinet (not on a drawer front panel). The elapsed time meter shall have a minimum read-out of 99999 hours and shall be as specified in MIL-M-7793.

3.3.7 Remote control. Signals for the remote control, indication and interlock shall be provided on a remote control panel for single unit and parallel operation. The remote control panel shall be connected to the unit by a type MSCU cable, as specified in MIL-C-915 and MIL-C-915/32. The cable will be furnished by the installation activity. The maximum length of the interconnection cable between the equipment and the remote control panel shall be 500 feet. The controls and indicators on the remote control panel shall be as follows:

(a) Pushbutton or switch for unit start-up and shut-down.
(b) Pushbutton or switch for output circuit breaker close and open.
(c) Indication light to show that the unit is energized.
(d) Indication light to show that the output circuit breaker is closed.
(e) Output current ammeter.

The frequency changer shall provide signals for 3.3.7 (a) and 3.3.7 (c) above. All controls and indicators as shown on figures 3 and 4 shall be located on the remote control panel. The installation activity will supply the remote control panel, complete with controls and indicators wired up, installed (including the interconnection cable) and ready for connection to the frequency changer.

3.3.8 Identification of equipment divisions. To provide identification that will aid documentation and to facilitate maintenance, each part, module, drawer or cabinet, and unit of the equipment shall be identified by reference designations in accordance with IEEE 200 and with 3.3.8.1 and 3.3.8.2.

3.3.8.1 Identification of parts leads. For each polarized part, such as electrolytic capacitors and rectifiers, the polarity identification shall be marked on the part mounting surface in order to help assure proper replacement of the part. For multiple lead parts such as transistors where proper circuit operation is dependent upon a specific lead hookup, appropriate lead identification shall be marked on the part mounting surface to help assure proper replacement of the part.

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3.3.8.2 Method of marking. Markings shall be permanent and legible. The markings on plastic or metallic materials shall be made by stamping, engraving, stenciling or rubber stamping with smudgeproof ink covered with a coat of clear lacquer or silk screening. Decalcomanias or paper labels shall not be used.

3.3.8.3 Identification plate. Equipment shall be equipped with an identification plate. The identification plate shall be metallic of types A, B, C or H as specified in MIL-P-15024 and MIL-P-15024/5.

3.3.8.4 Operation instruction plate. Equipment shall be equipped with an operation instruction plate. The plate shall be as specified in MIL-P-15024 and MIL-P-15024/5, types A, B, C or H (metallic only).

3.3.9 Human engineering. Human engineering effort shall be as specified in MIL-STD-1472.

3.3.10 Safety. The equipment shall be designed for the utmost safety of ship's personnel. The equipment shall be as specified in MIL-E-917. Personnel shall be protected from capacitor discharges as specified in requirement 1 of MIL-STD-454. When the equipment input circuit breaker (supplied by installation activity) is in the off position, the output circuit breaker (supplied by installation activity) and all circuits of the equipment shall be de-engergized (only the input terminals of the input circuit breaker and the output terminals of the output circuit breaker, and their associated controls shall be energized).

3.4 Component requirements.

3.4.1 Selection of parts. The contractor shall design the equipment such that parts, modules and drawers which perform similar functions will be standardized and interchangeable to the maximum extent practicable. The number of different parts to be designed into the equipment shall be kept to a minimum consistent with its intended use. Selection of parts shall be as specified in MIL-E-917. Request for the use of nonstandard parts shall be as specified in procedure I of MIL-STD-965. The use of nonstandard parts shall not be incorporated into the equipment design without prior approval from NAVSEA.

3.4.2 Contractor furnished parts. Supplementing other requirements for parts identification, contractor furnished parts shall be identified in parts lists and drawings by part manufacturer's name and part number in addition to the equipment manufacturer's part number.

3.4.3 Fuses. Fuses used in series with power semiconductors shall be the type for protection of semiconductor devices. Fuses shall be provided for power circuits and semiconductors as backup for the current limiting protection (see 3.2.7.4). Except for semiconductor protection, fuses shall be as specified in MIL-F-15160. Fuse voltage rating shall be not less than the maximum circuit application voltage.

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3.4.4 Solid state circuitry.

3.4.4.1 Thyristor firing circuit. Identical thyristor firing circuit modules shall be used throughout the equipment to the maximum extent possible. Firing circuits shall be designed for maximum noise immunity utilizing decoupling, shielding, low impedance, twisted pair cables, reverse bias and proper wiring layout to maximum extent possible.

3.4.4.2 Electrical control circuits. Established reliability components shall be used in the logic circuitry. Microcircuits shall be as specified in MIL-M-38510, class B or screened as specified in MIL-STD-883, level B-1 or B-2. Semiconductors shall be as specified in MIL-S-19500, level JANTX. Resistors and capacitors shall be as specified in applicable military specifications.

3.4.5 Semiconductor devices. Semiconductor devices shall be chosen and applied as specified in MIL-E-917 and as specified in 3.4.5.1 through 3.4.5.7. Unless otherwise specified, all semiconductor devices shall be silicon. The discrete semiconductors shall be as specified in TX requirements of MIL-S-19500 and the applicable detailed specification, where such semiconductors are available.

3.4.5.1 Standardization promotion. When a justifiable need exists for a device not covered by an existing military specification but which can be provided by extension of a series or family of military devices, such extension shall be employed in preference to choice of a device from a new family or series. The power semiconductor devices used (diodes and semiconductor control rectifier (SCR's)) shall be presently available without further development. The power semiconductors shall have at least two sources of supply, where possible.

3.4.5.2 Device selection prohibition. The equipment shall meet all requirements without the need for device selection beyond that covered by the applicable device detail specification. Semiconductor devices of the type designation shall be interchangeable for all equipment without any restrictions, including source of manufacture.

3.4.5.3 Reverse polarity prohibition. Devices of standard polarity shall be used wherever feasible. Devices of reverse polarity may be used only under the conditions applicable to the use of nonstandard parts (see 3.4.1).

3.4.5.4 Stress factors and limits. Circuits shall be designed and devices shall be applied so that the stress limits as specified in table VI will not be exceeded under any condition of operation, standby and during tests specified for the equipment. The values specified are limits and not design factors nor design margins. The equipment designer shall determine and provide sufficient margin between the applied stresses and the specified limits to make sure that the required life and reliability are achieved. In no case shall the relative values of the steady state voltage variation, over-voltage, transient voltage and surge voltage spikes, for the circuit under consideration, be considered to be less than the values specified in 3.2.1, 3.2.1.1 and 3.2.1.2 except where the design includes features to limit or suppress over-voltage and surge voltage to lower values. When such exception is used, the contractor shall provide a detail description of the design features and how the circuit functions to limit peak over-voltage and surge voltage impressed on the semiconductor devices to whatever lower values apply. Circuits which employ controlled rectifiers or transistors shall suppress surge voltages at least to not more than three times the

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supply nominal rms voltage. Semiconductor devices shall be applied in accordance with 3.4.5.5. Surge voltage suppressors shall be in accordance with 3.4.5.6. Heat dissipators (heat sinks, cooling fins) shall be in accordance with 3.4.5.7.

TABLE VI. Application stress limits and rerating factors.

Stress factor	Stress factor limit in relation rating factor		Rating factor
Semiconductor devices:	Lower	Upper	
(a) General:			
(1) Ambient temperature design			
a. Minimum T_{AL}	T_{AD}-50°C	T_{AD}-30°C	T_{AD} = equipment design ambient
b. Maximum T_{AH}	T_{AD}+15°C	T_{AD}+35°C	T_{AD} = design ambient temperature lower limit
(2) Ambient temperature measured			
a. Minimum, start-up	T_{AL}		
b. Maximum, operating	---	T_{AH}	T_{AH} = design ambient temperature higher limit
(b) Rectifiers and controlled rectifiers (SCR's):			
(1) Nominal working peak reverse voltage	---	0.5	Repetitive peak reverse voltage
(2) Repetitive peak reverse voltage	---	.7	Repetitive peak reverse voltage
(3) Nonrepetitive peak reverse voltage	---	1.0	Repetitive peak reverse voltage
(4) D.c. blocking voltage	---	0.4	Repetitive peak reverse voltage
(5) Nominal working peak forward voltage (SCR's)	---	.5	Forward blocking voltage
(6) Repetitive peak forward voltage (SCR's)	---	.7	Forward blocking voltage
(7) Nonrepetitive peak forward voltage (SCR's):			
a. Self-firing unacceptable	---	.8	Forward blocking voltage
b. Occasional self-firing acceptable	---	1.0	Forward blocking voltage
c. Same as b. and di/dt = 0.3 maximum allowable	---	1.0	Peak forward blocking voltage
(8) Reapplied forward blocking voltage (SCR's)	---	0.7	Forward blocking voltage at rated dv/dt, junction temperature and turn-off time

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TABLE VI. Application stress limits and rerating factors. - Continued

Stress factor	Stress factor limit in relation rating factor		Rating factor
	Lower	Upper	
Semiconductor devices:			
(b) Rectifiers and controlled rectifiers (SCR's) (*continued*):			
(9) Minimum duration of voltage reversal (SCR's)	2.0	---	Turn-off time
(10) Maximum rate of rise of forward blocking voltage (SCR's)	---	0.5	Minimum rate of rise, forward blocking voltage (dv/dt)
(11) Average forward current:			
a. Nominal value	---	.7	Forward current derated for conduction angle, wave form stud or case temperature T_C plus 15°C (ambient temperature T_A plus 15°C for lead-mounted devices) and duty cycle
b. Maximum value	---	1.0	
(12) Repetitive peak forward current:			
a. Nominal value	---	0.7	Repetitive peak forward current at maximum case temperature (maximum ambient temperature for lead-mounted devices)
b. Maximum value	---	1.0	
(13) Surge current:			
a. Peak, one cycle, nonrepetitive	---	0.7	Surge current, peak, half cycle (1/120 sec.) non-repetitive (log = logarithm to base 10, f = pulse frequency in Hz n = number of pulses, t = time in milliseconds
b. Effective peak and cycles duration	---	$0.2+0.28 \log \frac{f}{n}$	
c. Effective amplitude and duration sub-cycle	---	$1. - 0.6t$	
(14) Peak current during turn-on (SCR's)	---	0.7	Peak current during turn-on
(15) Maximum rate of current rise during turn-on (SCR's)	---	.5	Maximum rate of current rise during turn-on
(16) Peak forward gate current (SCR's)			
a. Maximum (for firing)	---	.7	Peak forward gate current
b. Minimum (for firing)	2.0	---	Forward gate current, minimum to cause firing
c. Maximum (during blocking)	---	0	Forward gate current, minimum to cause firing

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TABLE VI. Application stress limits and rerating factors. - Continued

Stress factor	Stress factor limit in relation rating factor		Rating factor
Semiconductor devices:	Lower	Upper	
(b) Rectifiers and controlled rectifiers (SCR's) (continued):			
(17) Peak forward gate voltage (SCR's)			
a. Maximum (for firing)	---	0.7	Peak forward gate voltage
b. Minimum (for firing)	2.0	---	Minimum firing voltage
c. Maximum (during blocking)	---	0	Minimum firing voltage
(18) Peak reverse gate voltage maximum (SCR's)	---	0.7	Peak reverse gate voltage
(19) Maximum gate power dissipation (SCR's):			
a. Average	---	.7	Average gate power dissipation
b. Peak	---	1.0	Peak gate power dissipation
(20) Junction temperature (calculated):			
a. Average	---	T_j (maximum)-40°C	T_j (maximum) = maximum junction temperature
b. Peak		T_j (maximum)-25°C	
(c) Voltage reference diodes:			
(1) Bias current	0.8	1.2	Zener test current (value for rated Zener voltage)
(2) Diode reverse current (Zener current):			
a. Minimum	0.1	---	Zener test current (value for rated Zener voltage)
b. Maximum	---	0.5	Maximum Zener current
c. Maximum, continuous duty	---	.3	Maximum Zener current
(3) Junction temperature (calculated)	---	100°C	

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TABLE VI. Application stress limits and rerating factors. - Continued

Stress factor	Stress factor limit in relation rating factor		Rating factor
	Lower	Upper	
Semiconductor devices:			
(d) Voltage regulator diodes:			
(1) Diode reverse current (Zener current):			
a. Minimum	0.1	---	Zener test current (value for rated Zener voltage)
b. Maximum (steady state)	---	0.5	Maximum Zener current
c. Maximum, continuous duty	---	.3	Maximum Zener current
d. Surge	---	.5	Surge current
(2) Junction temperature (calculated)			
a. Average (steady state)	---	T_j(maximum)-50°C	T_j (maximum) = maximum junction temperature
b. Peak	---	T_j(maximum)-25°C	
(e) Transistors:			
(1) Collector to emitter voltage	---	.5	Collector to emitter voltage, VCEO
(2) Collector current	---	.5	Maximum collector current
(3) Collector power dissipation:			
a. Maximum (steady state average)	---	.5	Collector power dissipation, Pc, derated for T_C + T (or T_A + T) where T=50°C for silicon or 15°C for germanium
(4) Junction temperature (calculated)			
a. Average (steady state)	---	T_j(maximum)-50°C	T_j (maximum) = maximum junction temperature
b. Peak	---	T_j(maximum)-25°C	

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3.4.5.5 Application limitations. The following limitations and design guides shall apply in the application of semiconductor devices:

(a) Devices designed and rated for conduction cooling shall be mounted on metallic heat dissipators. Due regard shall be given to the compatibility of materials and to the finish coatings on parts mounted together in an assembly. Particular attention shall be given to the design of connections and joints in both electrical and thermal circuits to make sure that their effectiveness will be maintained for the life of the equipment under adverse shipboard conditions, including shock, vibration, thermal cycling, high and low humidity and salt-laden atmosphere. Means for maintaining positive contact pressure on bolted or clamped joints, under all service conditions, taking into account the aging (including creep, cold flow, relaxation, fatigue, shrinkage and so forth) of materials, shall be included.

(b) Direct mounting of devices to their heat dissipator with electrical insulation of the heat dissipators, as necessary, is preferred to the use of intervening electrical insulating material between the devices and their heat dissipators. This preferred arrangement shall be used except where it is impracticable to insulate the heat dissipators, or where other technical considerations make electrical insulation of the semiconductor device from its mounting surface necessary.

(c) Semiconductor devices shall not be operated in parallel to provide current capacity.

(d) Semiconductor devices (diodes, rectifiers, SCR's and transistors) shall not be operated in series. Transistors shall not be used in series to obtain higher voltage capacity.

(e) Voltage reference diodes and voltage regulator diodes shall not be used in parallel to obtain higher current capacity.

3.4.5.6 Surge voltage suppression. Circuits containing devices which do not have inherent capability of withstanding voltage surges of the level as specified in 3.4.5.4 without exceeding allowable stresses shall be protected by incorporating adequate surge voltage suppression within the circuit. The internal generation of voltage surges shall be minimized with respect to both frequency of occurrence and amplitude. Means shall be provided within the circuit both for suppressing any remaining voltage surges generated within the circuit that would otherwise be excessive and for preventing external propagation at a peak level greater than three times the rms voltage for the a.c. output circuit and that permitted as specified in 3.2.1, 3.2.1.1 and 3.2.1.2 for the input circuit. Surge voltage suppressors shall be in accordance with MIL-E-917.

3.4.5.7 Heat dissipators. Heat dissipators for semiconductor devices shall be constructed of metallic material which is either corrosion-resistant, or which is treated or coated to resist corrosion under shipboard environmental conditions. Surfaces which contact the semiconductor device, and surfaces to which electrical contacts are made shall not be painted or anodized, but shall be plated or otherwise coated to form surfaces to which the connections or junctions are made and which will retain their mechanical, thermal and electrical effectiveness for the life of the equipment. The method of processing and the preparation of the connections and joints shall be as specified on the applicable heat dissipator drawing. Heat dissipators shall be sized to dissipate the heat generated by the

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semiconductor device of highest losses that may be expected to conform to the applicable semiconductor device specification, and to do so under the highest ambient temperature conditions without exceeding the limiting temperatures as specified in table VI. Maximum power dissipation may be estimated from calculations taking into account the mode of operation of the device in the circuit, the worst stress levels applied and the limiting device characteristics as covered by the device specification. Maximum power dissipation may also be estimated by using published data for a particular brand of device, and scaling up the losses using a factor calculated from parameter limits applicable to the particular brand of device and specification limits, taking into account the influence of the parameters considered on the losses under the mode of operation in the circuit and allowing for any difference in form factor involved in the way the values are given. Where maximum power dissipation at rated conditions is specified and controlled by the device specification, this value corrected for the stress levels of the application, shall be used in the design.

3.5 Environmental requirements.

3.5.1 Shock. The equipment shall withstand the heavyweight high-impact shock test in accordance with MIL-S-901 for grade A, class I, type A, deck-mounted equipment. The equipment shall be hard-mounted, without the use of resilient mounts. The equipment shall operate in parallel with another converter and continually carry load, with the output as specified in DOD-STD-1399, section 300, type III limits during shock test.

3.5.2 Vibration. Equipment shall meet the type I vibration requirements as specified in MIL-STD-167-1 up to and including 15 Hz without sway braces. When a higher frequency up to and including 25 Hz is required for a class of ship (see MIL-STD-167-1), the equipment shall meet these requirements with sway braces.

3.5.3 Noise. Equipment shall be solidly mounted to the ship structure and shall be designed and tested to meet the airborne and structureborne noise requirements as specified in MIL-STD-740. The limits of airborne noise shall be grade B. The limits of structureborne noise shall be as specified on figure 5.

3.5.4 Electromagnetic interference emission. Equipment shall conform to the electromagnetic interference and susceptibility limits of MIL-STD-461, part 5, CE03, RE02 and RS01.

3.5.5 Ambient temperature. Equipment shall start and operate satisfactorily and conform to all requirements specified herein throughout an ambient temperature range of 0 to 50°C.

3.5.6 Cooling.

3.5.6.1 Cooling air. The cooling air for the air cooled units shall be ambient air of a maximum temperature of 50°C.

3.5.6.2 Cooling system failure protection. Protection of the equipment against cooling failure shall be provided by means of automatic de-energizing of the equipment. The input circuit breaker providing power to the equipment shall be shunt tripped in response to overtemperature of equipment components and internal air temperature. Fan failure shall trigger an alarm but shall not trigger shut down.

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3.5.7 Humidity conditions. Equipment shall operate with relative humidities ranging up to 95 ± 5 percent by both intermittent and continuous periods, including conditions wherein condensation takes place in and on the equipment cabinets, in the form of both water and frost.

3.5.8 Clearance for ventilation and maintenance. Clearance required between the equipment and adjacent equipment or structure shall be not greater than specified in table VII.

TABLE VII. Clearance for ventilation and maintenance.

Bottom	none
Sides or ends	18 inches
Rear	6 inches
Top	6 inches
Front	30 inches

3.5.9 Ship motion. Equipment shall operate when inclined 15 degrees permanently in any direction and when installed on board ships having the following pitch and roll:

	Magnitude	Period
Pitch	10 degrees	5 seconds
Roll	45 degrees	8 seconds

3.5.10 Lifting provision. Each cabinet shall have removable eye-bolts provided on top to permit hoisting and handling. The eye-bolts shall be located to balance the weight of the cabinet as prepared for shipping in the upright position. The lifting attachment shall be of sufficient capacity to carry at least five times the total weight of the cabinet. The cabinet shall remain drip-proof when the eye-bolts are removed.

3.6 Reliability. The reliability measure shall be mean-time-between-failures (MTBF) in accordance with MIL-STD-781 and shall be not less than 13,000 hours.

3.6.1 Definition of failure. The definition of failure for the purpose of determining the achievement of reliability characteristics shall be:

(a) The input and output performance not in accordance with requirements specified herein.
(b) Malfunction requiring shutdown because of damage or incipient damage.

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3.6.2 Reliability analysis. The reliability analysis shall employ one or more of the prediction methods as specified in MIL-HDBK-217. Failure rates shall be selected as specified in MIL-HDBK-217. When component failure rates are not available as specified in MIL-HDBK-217, failure rates from other sources shall be used. Failure rates from other sources shall be fully documented and substantiated. For the reliability analysis, the failure rate of each component shall be calculated based on maximum ambient temperature in which the component operates in the equipment. The ambient temperature at each component shall be calculated from the input cooling air temperature which is 50°C. The environmental factor shall be naval unsheltered as specified in MIL-HDBK-217.

3.6.3 Failure modes and effects analysis (FMEA). The FMEA shall be prepared in accordance with task 101 of MIL-STD-1629.

3.7 Maintainability. The mean time to repair shall be not more than 2 hours. The maximum time to repair shall be not more than 6 hours.

3.7.1 Test equipment. The equipment shall require the use of only the following test equipment:

(a) Clamp-on volt ammeter, 0 to 600 amperes and 0 to 600 volts, a.c.; Nomenclature: CV-633 or AN/USM-33.
(b) Multimeter, 0 to 5000 volts, direct current (d.c.), 0 to 1000 volts, a.c.; 0 to 10 amperes, d.c. and 0 to 30 megohm range; Nomenclature: AN/PSM-4.
(c) Oscilloscope, X-Y, low frequency; Nomenclature: AN/USM-368 (Hewlett Packard Model 130C or equivalent).
(d) D.c. milliameter, 0 to 150 ma. range; Nomenclature: CV-931-4904003.
(e) A.c. voltmeter, 0 to 600 volts range type MR71W600ADVV in accordance with MIL-M-16034 and MIL-M-16034/9.
(f) Portable tester (see 3.7.2).

Test equipment other than listed herein may be used only under the conditions applicable to the use of nonstandard parts (see 3.4.1). Test equipment required for maintenance of the equipment and not listed herein or in the NAVELEX Coordinated Shipboard Allowance List, Portable Testing and Measuring Equipment shall be furnished by the contractor with the equipment.

3.7.2 Portable tester. A portable tester, Teledyne Inet Part Number 10008860, will perform functional testing of the logic circuits and the power switching drawers at the organizational level.

3.7.3 Interchangeability. In no case shall parts be physically interchangeable or reversible unless such parts are also interchangeable or reversible with regard to function, performance and strength.

3.8 Drawings. When specified (see 6.2.1), drawings shall be prepared in accordance with the data ordering document included in the contract or order (see 6.2.2).

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3.9 Workmanship. Particular attention shall be given to neatness and thoroughness of soldering, marking of parts and subassemblies, wiring, welding and brazing, plating, riveting, finishes, machine operations, screw assemblies and freedom of parts from burrs and sharp edges.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Calibration system requirements. The contractor shall provide and maintain a calibration system for test equipment in accordance with the data ordering document (see 6.2.2).

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) First article inspection (see 4.4).
(b) Quality conformance inspection (see 4.5).

4.3 Test conditions. Unless otherwise specified herein, tests shall be conducted under nominal ambient temperature of 25°C. The specific ambient temperature at the time of test shall be measured and recorded. Transient voltage or current traces taken by oscillograph or pictures of oscilloscope shall have a peak-to-peak magnitude, and speed so that the variable can be clearly measured.

4.4 First article inspection. When specified (see 6.2.1), one unit of each power rating shall be subjected to the examination and test as specified in table VIII.

TABLE VIII. First article inspection.

Examination and tests	Requirement paragraph	Inspection paragraph
General examination	3.3	4.6 and 4.7
Insulation resistance	3.3.2.3	4.7.2 and 4.7.3
Dielectric withstanding voltage[1/]	3.3.2.3	4.7.4
Self-protection from input voltage variation and transient	3.2.1.1	4.7.6
Input current harmonics	3.2.1.3	4.7.7
Starting current and starting time	3.2.1.5, 3.2.1.5.1 and 3.2.1.6	4.7.8
Voltage and frequency regulation	3.2.2.1 and 3.2.2.5	4.7.9

See footnote at end of table.

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TABLE VIII. First article inspection. - Continued

Examination and tests	Requirement paragraph	Inspection paragraph
Line drop compensation	3.2.2.2	4.7.10
Voltage adjustment	3.2.2.3	4.7.11
Transient voltage and frequency deviation and recovery	3.2.2.4 and 3.2.2.6	4.7.12
Amplitude and frequency modulation	3.2.2.7	4.7.13
Output voltage waveform	3.2.2.8	4.7.14
Output phase displacement angle	3.2.2.9	4.7.15
Output voltage balance under balanced load	3.2.2.10	4.7.16
Output voltage balance under unbalanced load conditions	3.2.2.11	4.7.17
Protection for utilization equipment	3.2.7.5	4.7.18
Input voltage change effect	3.2.2.12	4.7.19
Efficiency and no-load input current	3.2.1.7, 3.2.4 and 3.2.5	4.7.20
Power rating	3.2.3 and 3.2.3.1	4.7.21
Input current balance	3.2.1.4	4.7.22
Overload	3.2.6	4.7.23
Short circuit and fault current	3.2.7.2 and 3.2.7.3	4.7.24
Parallel operation	3.2.8, 3.2.8.1, 3.2.8.2 and 3.2.8.3	4.7.25
Shock	3.5.1	4.7.26
Vibration	3.5.2	4.7.27
Noise (structureborne)	3.5.3	4.7.28
Noise (airborne)	3.5.3	4.7.29
Cooling system failure protection	3.5.6.2	4.7.30
Electromagnetic interference emission and susceptibility	3.5.4	4.7.31
Temperature rise and high temperature operation	3.5.5	4.7.32
Cooling	3.5.6 and 3.5.6.1	4.7.32
Inclined operation	3.5.9	4.7.33
Grounding	3.2.1.8	4.7.34
Humidity	3.5.7	4.7.35
Remote start-up and shutdown	3.3.7	4.7.36
Stress level conformation	3.4.5.4 and 3.6.2	4.7.37
Maintainability demonstration	3.7	4.7.38
Reliability 1/	3.5	4.7.39

1/ Any of these tests may be conducted concurrently except that the dielectric withstanding voltage test shall be conducted after all other tests, but before the reliability tests have been completed. The reliability tests shall be conducted only after satisfactory completion of all other tests.

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4.4.1 First article inspection report. A first article inspection report shall be submitted in accordance with the data ordering documents included in the contract or order (see 6.2.2).

4.5 Quality conformance inspection. Each unit offered for delivery shall be subjected to the quality conformance inspection specified in table IX.

TABLE IX. Quality conformance inspection.1/

Examination and tests	Requirement paragraph	Inspection paragraph
General examination	3.3	4.6 and 4.7
Insulation resistance	3.3.2.3	4.7.2 and 4.7.3
Self-protection from input voltage variation and transient	3.2.1.1	4.7.6
Input current harmonics	3.2.1.3	4.7.7
Voltage and frequency regulation	3.2.2.1 and 3.2.2.5	4.7.9
Line drop compensation	3.2.2.2	4.7.10
Voltage adjustment	3.2.2.3	4.7.11
Transient voltage and frequency deviation and recovery	3.2.2.4 and 3.2.2.6	4.7.12
Amplitude and frequency modulation	3.2.2.7	4.7.13
Output voltage waveform	3.2.2.8	4.7.14
Output voltage balance under balanced load	3.2.2.10	4.7.16
Output voltage balance under unbalanced load	3.2.2.11	4.7.17
Overload	3.2.6	4.7.23
Short circuit and fault current	3.2.7.2 and 3.2.7.3	4.7.24
Parallel operation	3.2.8, 3.2.8.1, 3.2.8.2 and 3.2.8.3	4.7.25
Endurance run	3.2.3 and 3.5	4.7.41

1/ Any of these tests in table IX may be conducted concurrently except that the endurance run shall be conducted after all other tests have been completed, to verify that all requirements have been met.

4.5.1 Test reports. The contractor shall prepare quality conformance test reports in accordance with the data ordering document (see 6.2.2).

4.6 General examination. Parts on assemblies and completed units shall be examined as specified in 4.6.1 and 4.6.2.

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4.6.1 Major parts. Examination shall be conducted to determined that:

(a) The following items conform to the applicable specification and that the size, type and rating are as shown on the accepted drawing:

(1) Resistors and rheostats.
(2) Instruments, instrument transformers, shunts and other associated accessories.
(3) Capacitors.
(4) Switches.
(5) Semiconductor devices.
(6) Terminal boards.
(7) Laminated materials, such as glass mica glass (GMG), glass silicon glass (GSG); used for terminal boards, sub-panels and so forth.

(b) Materials used in the construction of wound components comply with the material requirements as shown on the accepted drawing.

(1) Windings were given varnish impregnation in accordance with the procedures shown on the accepted drawing.
(2) The resistance of all windings is as shown on the accepted drawing.

4.6.2 Completed equipment. Examination of the completed equipment shall be conducted to determine that:

(a) Parts, modules and drawers are mounted as shown on the accepted drawing.
(b) Parts and modules which require servicing, repair, replacement or periodic adjustment during the life of the equipment are readily accessible.
(c) Parts, modules and drawers are marked with identifying symbols and these symbols agree with those used on the accepted drawing.
(d) Terminals of parts and modules and terminal boards are marked with identifying letters, numbers or both, and this identification agrees with those used on the accepted drawing.
(e) Wire used for interconnecting parts, modules and drawers are of the type shown on the accepted drawing.
(f) Wiring is neatly formed into groups and laced with flame-resistant tie or glass cord and supported or clamped in a manner which will prevent chafing of the insulation due to vibration and shock.
(g) Cable clamps used are of flame-resistant material.
(h) There are no splices in any of the wires.
(i) Wire groups from hinged doors and panels are formed and clamped in a manner that sharp bends do not occur with the panel or door in the open or closed position.
(j) Sufficient slack in wiring is allowed so that the weight of the harness is not supported by the terminal connections; so that at least two replacements can be made if lugs are clipped off at the wire end.
(k) Cable entrance provisions and connections are as shown on the accepted drawing.

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(l) Sufficient space is provided for these connections and sufficient cable support within the drawers and cabinets, and connections made at the appropriate terminals.
(m) Wherever wires run through holes in metal partitions or chassis, grommets are provided for mechanical protection.
(n) Wires are not bent around sharp corners which may injure the insulation.
(o) Wires are connected by either bolted or soldered connections.
(p) Bolted connections are provided with locking devices.
(q) Both ends of all wires are marked with designations as shown on the accepted drawing.
(r) The equipment pull-out drawers have stops to lock in pull-out position.
(s) Nonidentical drawers cannot be interchanged in the equipment cabinet.
(t) Identification and instruction plates are furnished as shown on the accepted drawing.

4.7 Tests.

4.7.1 Size and weight. Equipment shall be measured and weighed with dimensions and weight recorded.

4.7.2 Creepage and clearance. Creepage and clearance distances shall be demonstrated by actual measurement when it appears close.

4.7.3 Insulation resistance tests. Insulation resistance test shall be conducted in accordance with method 302 of MIL-STD-202, test condition B.

4.7.4 Dielectric withstanding voltage tests. Dielectric withstanding voltage tests shall be conducted in accordance with method 301 of MIL-STD-202. The rms test voltage shall be twice rated circuit voltage plus 1000 V. The duration of test shall be 60 seconds. Any evidence of arcing, flashover or punctured insulation shall be interpreted as a failure of the test. Corona (visible, audible or odorous) shall not be considered as a failure.

4.7.5 Enclosures. Acceptability of the enclosure shall be determined in accordance with MIL-STD-108.

4.7.6 Self-protection from input voltage variation and transients. The test shall be performed as follows:

(a) Equipment shall be operated with nominal input voltage at no-load and rated-load. The input voltage to the equipment shall be raised and lowered beyond the specified operation range for 10 ms. The input voltage, output voltage and output current shall be recorded with an oscillograph.
(b) Equipment shall be operated at nominal input voltage to supply rated-load. Interrupt the input power for a period of 30 seconds and then restore to the original condition. The equipment shall be restarted manually. The equipment input voltage, output voltage and output current shall be recorded with an oscillograph.

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4.7.7 Input current harmonics. The equipment shall be operated at nominal input voltage and frequency. The input current harmonics of each input line shall be determined at no load, 50 percent load and rated-load. Harmonics less than 0.1 percent may be disregarded. The current harmonics shall be measured by means of shunts connected in each input lines to the equipment. The fundamental and harmonics of the voltage drops across the shunts shall be determined by a harmonic analyzer.

4.7.8 Starting current and starting time. The nominal voltage shall be connected to the equipment input, the equipment shall be started with no-load at the equipment output. The input starting current and output voltage shall be measured with an oscillograph. The test shall be repeated with full load connected to the equipment. The starting time and starting current shall be determined. Attempt to energize the equipment when the input voltage phase rotation is in the reverse order (AC-CB-BA). Note any damage to the equipment.

4.7.9 Voltage and frequency regulation. For first article test, the equipment shall be connected to an input, whose voltage and frequency can be varied as specified in table X. After the equipment has been stabilized at load, the output steady state voltage and frequency regulation shall be determined for test conditions 1, 2, 3, 7, 9, 11, 23, 25, and 27 of table X. For quality conformance test, the output steady state voltage and frequency regulation shall be determined for the following test conditions as specified in table X: 2, 8, 9, 10, 13, 14, 15, 24, 25, 26, 29, 30, 31, 36 and 37.

4.7.10 Line drop compensation. The unit shall be connected to rated load and the load compensation potentiometers R1A, R1B, and R1C shall be adjusted to obtain a 7 percent boost (31.5 V) in output voltage. The load shall be reduced to half load and to no load with no change in potentiometer setting. The average of the three output phase voltages shall vary linearly from full load to no load within plus or minus 0.5 percent (plus or minus 2.25 volts). This test shall be performed at nominal input conditions.

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TABLE X. Conditions of input and output.

Test condition numbers	Input						Output (of rated voltage and frequency)							
	Voltage within the specified range			Frequency within the specified range			Rated load				Power factor			
	max	nom	min	max	nom	min	0	50 percent	100 percent	125 percent	0.6 lag.	0.9 lag.	1.0	0.9 leading
1	X	-	-	X	-	-	X	-	-	-	-	-	-	-
2	-	X	-	-	X	-	X	-	-	-	-	-	-	-
3	-	-	X	-	-	X	X	-	-	-	-	-	-	-
4	X	-	-	X	-	-	-	X	-	-	X	-	-	-
5	-	X	-	-	X	-	-	X	-	-	X	-	-	-
6	-	-	X	-	-	X	-	X	-	-	X	-	-	-
7	X	-	-	X	-	-	-	X	-	-	-	X	-	-
8	X	-	-	-	X	-	-	X	-	-	-	X	-	-
9	-	X	-	-	X	-	-	X	-	-	-	X	-	-
10	-	-	X	-	X	-	-	X	-	-	-	X	-	-
11	-	-	X	-	-	X	-	X	-	-	-	X	-	-
12	X	-	-	X	-	-	-	X	-	-	-	-	X	-
13	X	-	-	-	X	-	-	X	-	-	-	-	X	-
14	-	X	-	-	X	-	-	X	-	-	-	-	X	-
15	-	-	X	-	X	-	-	X	-	-	-	-	X	-
16	-	-	X	-	-	X	-	X	-	-	-	-	X	-
17	X	-	-	X	-	-	-	X	-	-	-	-	-	X
18	-	X	-	-	X	-	-	X	-	-	-	-	-	X
19	-	-	X	-	-	X	-	X	-	-	-	-	-	X
20	X	-	-	X	-	-	-	-	X	-	X	-	-	-
21	-	X	-	-	X	-	-	-	X	-	X	-	-	-
22	-	-	X	-	-	X	-	-	X	-	X	-	-	-
23	X	-	-	X	-	-	-	-	X	-	-	X	-	-
24	X	-	-	-	X	-	-	-	X	-	-	X	-	-

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TABLE X. Conditions of input and output. - Continued

Test condition numbers	Input						Output (of rated voltage and frequency)							
	Voltage within the specified range			Frequency within the specified range			Rated load				Power factor			
	max	nom	min	max	nom	min	0	50 percent	100 percent	125 percent	0.6 lag.	0.9 lag.	1.0	0.9 leading
25	-	X	-	-	X	-	-	-	X	-	-	X	-	-
26	-	-	X	-	X	-	-	-	X	-	-	X	-	-
27	-	-	X	-	-	X	-	-	X	-	-	X	-	-
28	X	-	-	X	-	-	-	-	X	-	-	-	X	-
29	X	-	-	-	X	-	-	-	X	-	-	-	X	-
30	-	X	-	-	X	-	-	-	X	-	-	-	X	-
31	-	-	X	-	X	-	-	-	X	-	-	-	X	-
32	-	-	X	-	-	X	-	-	X	-	-	-	X	-
33	X	-	-	X	-	-	-	-	X	-	-	-	X	-
34	-	-	X	-	-	X	-	-	X	-	-	-	X	-
35	-	X	-	-	X	-	-	-	-	X	X	-	-	-
36	-	X	-	-	X	-	-	-	-	X	-	X	-	-
37	-	X	-	-	X	-	-	-	-	X	-	-	X	-

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4.7.11 Voltage adjustment. After the equipment has been stabilized at rated-load, the output voltages at the equipment output terminals shall be adjusted over their full ranges, using the voltage adjustment devices. For first article test, the test shall be conducted for test conditions 2, 24, 25, and 26 as specified in table X. For quality conformance test, the test shall be conducted for test conditions 2, 8, 9, 14, 25, 30, 31 and 36 as specified in table X.

4.7.12 Transient voltage and frequency deviation and recovery. With the equipment operating with nominal input, a balanced 3-phase load of 0.9 power factor shall be changed in one step for the following conditions:

(a) From no-load to half-load
(b) From half-load to rated-load
(c) From rated-load to half-load
(d) From three-fourths load to one-fourth load

Oscilloscope pictures shall be taken of the load application and removal, to determine the voltage transient and recovery time. This test shall be conducted at nominal input conditions. The frequency transient and recovery test shall be performed the same way except that load application and removal shall be from no-load to rated-load and from rated-load to no-load in one step, respectively.

4.7.13 Amplitude and frequency modulation. For first article test, amplitude and frequency modulation of the equipment shall be determined at the equipment output terminals for test conditions 2, 9, 14, 25, and 30, as specified in table X. The voltage amplitude shall be measured with an oscilloscope. The oscilloscope settings shall be 10 V per division and 10 ms or 20 ms per division. Oscilloscope pictures shall be taken. For quality conformance test, the test conditions of 2, 9, 14, 25 and 30 as specified in table X, shall be performed.

4.7.14 Output voltage waveform. The harmonic content and output voltage spikes shall be determined. For first article, this test shall be performed with the equipment operating under the test conditions 2 and 25, as specified in table X. For quality conformance test, this test shall be performed under the test conditions 2, 9, 14, 25 and 30 as specified in table X.

4.7.15 Output phase displacement angle. The equipment output displacement angle shall be measured at the output terminals. For first article test, this test shall be performed for test conditions 2 and 25, as specified in table X. For quality conformance test, this test shall be performed for test conditions 2, 9, 14, 25 and 30 as specified in table X.

4.7.16 Output voltage balance under balanced load. After the equipment has been stabilized at rated-load, the output voltage of each phase at the equipment output terminal shall be recorded. For first article test, this test shall be performed for test conditions 1, 2, 3, 7, 9, 11, 23, 25, and 27, as specified in table X. For quality conformance test, this test shall be performed for test conditions 5, 26 and 35 as specified in table X, all at 0.9 lagging power factor load.

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4.7.17 Output voltage balance under unbalanced load. Equipment shall be operated with nominal input, apply a 3-phase, 0.6 power factor load across the equipment output phases to have the lines carry current, of 75 percent of rated current. A single phase 0.6 power factor load shall be applied across the equipment output phase A-B to have line A carry an additional current of 15 percent of rated current. The output voltage of all phases shall be recorded at the equipment output terminals. The test shall also be conducted using a single phase 0.6 power factor load across equipment output phase B-C and the C-A. The equipment shall be operated at nominal input. A single phase, 0.6 power factor load shall be applied across equipment output phases A-B to have line A carry a current of 15 percent of the rated current. The output voltages of all phases shall be recorded. The test shall also be conducted using a single phase 0.6 power factor load across equipment output phases B-C and then C-A. The above tests shall be repeated for 0.9 power factor load.

4.7.18 Protection for utilization equipment. Equipment shall be operated at nominal input at no-load. Voltage signals shall be injected into the equipment sensing circuits to simulate the condition of equipment output overvoltage and undervoltage, as specified in table IV. The output voltage and the time required to shut down the equipment in the case of overvoltage shall be recorded by oscilloscope or oscillograph.

4.7.19 Input voltage change effect. Equipment shall be operated within the input specified voltage and frequency ranges, the input voltage shall be increased or decreased to the specified amount (see 3.2.2.12) instantaneously. The input and output voltages shall be recorded by oscillograph to determine the voltage transient and recovery time. The test shall be performed at no-load and rated-load, 0.9 power factor.

4.7.20 Efficiency and no-load input current. The equipment shall be loaded with 0.9 power factor loads for this test. Meters shall be inserted in the input and output for reading voltage, current and wattage. For first article test, readings shall be made for test conditions 2, 25, 30 and 31 as specified in table X. For quality conformance test, readings shall be made for test conditions 2, 9 and 25 as specified in table X. Efficiency shall be calculated by means of the following formula:

$$\text{Overall percent efficiency} = \frac{\text{Output power}}{\text{Input power}} \times 100$$

4.7.21 Rating. The equipment shall be operated at nominal input for 2 hours each at the following loads:

(a) Rated-load, 0.6 power factor lagging.
(b) Rated-load, 0.9 power factor lagging.
(c) Rated-load, unity power factor.
(d) Rated-load, 0.9 power factor leading.

Input power, output power, output current, output frequency and output voltage shall be recorded at the beginning and at the end of each 2-hour load run.

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4.7.22 Input current balance. The input voltage to the equipment shall be nominal with 3 percent unbalance among the three phases. The equipment shall be operated at 10 percent load, 50 percent load, and rated-load at rated output voltage and frequency. The input current of each line to the equipment shall be measured with indicating meter having initial accuracy of 1 percent or better. This test shall be repeated with the input 3-phase voltages set at the nominal value and balanced within 4 V.

4.7.23 Overload. The equipment shall be operated at nominal input and rated output, and then overloaded to 125 percent of rated-load at rated output voltage, 0.9 power factor for more than 1 minute. The output voltages, output currents and output amplitude modulation shall be determined by an oscillograph.

4.7.24 Short circuit and fault current. A 3-phase and single phase short circuit fault shall be applied to the equipment output terminals. This test shall be conducted with input voltage of maximum, nominal and minimum in the specified range. The fault current and the time required to shutdown the equipment shall be recorded with an oscillograph and also with an oscilloscope.

4.7.25 Parallel operation. For first article test, three units of the same kW rating shall be operated in parallel. For quality conformance test, two units of the same kW rating shall be operated in parallel. When three units are operated, two of the units shall be supplied from one input power source and the other from a different input power source. After the units have been automatically synchronized to operate in parallel, no further adjustment shall be made for the remainder of this test. The total load of the paralleled units shall be varied from no-load to 90 percent of the combined kW and kVA rating of the units at 0.9 power factor in 25 percent load steps and back to no-load in the same manner. The units shall operate for at least 1 hour at each load step. The difference in kW and kVA supplied by each unit to the load shall be recorded and computed. For each load step the parallel test shall be conducted at the maximum, nominal and minimum input voltage for 20 minutes each in the specified range. Simulated internal fault or door interlock opening shall be applied to one of the paralleled units to demonstrate selective typing; the equipment output voltage shall be recorded by an oscillograph or oscilloscope photo. After completion of the selective tripping test, the units shall operate in parallel at 20 percent load for 16 hours continuously.

4.7.26 Shock. Two frequency changers shall be connected in parallel output and, while operating, subjected to high-impact heavyweight shock tests in accordance with MIL-S-901 and as specified in 4.7.26.1.

4.7.26.1 Test features. Test features shall be as follows:

(a) Type A (for principal units), deck-mounted equipment.
(b) Grade: A.
(c) Equipment class: 1.
(d) Load applied: 20 percent of rated-load, both units.
(e) Mounting: Hard-mounting.
(f) Prior to and following the shock test, the units shall be tested as specified in DOD-STD-1399, section 300, type III limits.
(g) Definition of failure to perform principal functions shall be as follows:

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(1) Breaking of any parts, including mounting bolts. Minor chipping of parts such as plastic knobs and cases and minor distortion of parts will be permitted where such chipping or distortion cannot in any manner impair operation of the equipment as specified.
(2) Appreciable distortion of any parts, including enclosure and framework.
(3) A value of insulation resistance lower than that permitted by this specification (see 3.3.2.2).
(4) Failure to pass visual examination. The equipment shall be carefully examined after removing all removable panels, doors and drawers to ascertain any mechanical damage. If necessary, partial disassembly shall be performed to aid in determining any possible damage.
(5) Failure to continually carry rated-load during shock test and to perform electrical test following shock. No adjustment or replacement of damaged parts shall be permitted after the shock test and prior to the tests specified in 4.7.26.1(h).

(h) On completion of the shock tests the units shall be tested for insulation resistance and output performance as specified in DOD-STD-1399, section 300, type III limits.

4.7.27 Vibration. Vibration tests shall be conducted in accordance with the type I vibration requirements of MIL-STD-167-1 up to and including 15 Hz. The equipment shall be tested with 10 percent or more rated-load. Sway braces shall not be used unless the required frequency range is up to and including 25 Hz.

4.7.27.1 Test feature. The definition of failure as specified in 4.7.26(g) shall also apply to vibration tests.

4.7.28 Structureborne noise. The structureborne noise shall be determined in accordance with MIL-STD-740. For the noise tests, the equipment shall be operated at full load. Measurements shall be made in three orthogonal directions above every isolation mount. The power sum is defined as:

$$(\text{Power sum adb})_{\kappa} = \left[\left(10 \log_{10} \frac{1}{n} \sum_{i=1}^{n} 10^{(a_i/10)}\right) \right]_{\kappa}$$

Where:

a_i = acceleration level in adb (referred to 10^{-3} cm/sec^2) above the ith isolation mount.

n = the number of measurement points in a given direction (no. of isolation mounts).

κ = direction - vertical, longitudinal or transverse.

The power sum as measured for any of the three directions shall not exceed the limits of figure 5. The contractor shall prepare test reports in accordance with the data ordering document (see 6.2.2).

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4.7.29 Airborne noise. The airborne noise shall be determined in accordance with MIL-STD-740. Load conditions during tests and reports of tests shall be the same as specified for structureborne noise tests (see 4.7.28).

4.7.30 Cooling system failure protection test. The equipment shall be operated at nominal voltage input and at rated-load. The cooling fan circuits shall be interrupted to simulate fan failures. The equipment input voltage and output voltage shall be observed with an oscilloscope.

4.7.31 Electromagnetic interference (EMI) emission and susceptibility. An electromagnetic interference emission and susceptibility test shall be conducted in accordance with MIL-STD-462. Tests shall be conducted on each input and output conductor of the equipment. Data for both the broadband and narrowband of the conducted emission levels shall be taken and incorporated in the EMI test report in accordance with the data ordering document (see 6.2.2).

4.7.32 Temperature rise and high temperature operation. Temperature rise tests shall be conducted at an ambient temperature of 50°C. The test methods to be employed and the precautions to be observed shall be as specified in 4.7.32.1 through 4.7.32.5.

4.7.32.1 Assembly and mounting of equipment. Temperature rise tests shall be made only with the equipment completely assembled and mounted in the manner for which it is designed. Barriers shall be placed adjacent to the enclosure to simulate shipboard space restrictions. The minimum temperature measurement points shall be as follows:

(a) Power semiconductor.
(b) Capacitors.
(c) Power transformer and reactor coil windings.
(d) Ambient temperature.
(e) Cooling air inlet and outlet.

4.7.32.2 Method of temperature measurements. The methods of temperature measurement of ambient and parts or subassemblies shall be in accordance with MIL-E-917 except that thermometer shall not be used.

4.7.32.3 Method of loading. The equipment shall be operated at test conditions 23, 25 and 27 as specified in table X. The temperatures shall reach the steady final value as specified in 4.7.32.4 for each input voltage and for each load condition. Temperature rises shall be in accordance with MIL-E-917.

4.7.32.4 Duration of test. The consecutive temperature measurement of each part or module shall be recorded not more than 3 minutes apart. The temperature rise test shall be continued until the temperature rise of parts or modules that can be observed or recorded during test, have attained a steady final value. Steady final value is that value of temperature which will remain at a 2°C band for a period of 2 hours, while the ambient temperature is maintained at 50°C.

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4.7.32.5 High temperature operation. After the equipment has attained a steady final value as specified in 4.7.32.4, the following tests shall be conducted at the ambient temperature of 50°C.

(a) Voltage and frequency regulations test as specified in 4.7.9, at test conditions 2, 7, 9, 11, 14, 23, 25, 27, and 30 of table X.
(b) Transient voltage and frequency deviation and recovery test as specified in 4.7.12.
(c) Amplitude and frequency modulation test as specified in 4.7.13, at the conditions 2, 7, 9, 11, 14, 23, 25, 27, and 30 of table X.
(d) Output voltage waveform test as specified in 4.7.14, at test conditions 2, 7, 9, 11, 14, 23, 25, 27, and 30 of table X.
(e) Short circuit protection test as specified in 4.7.24.
(f) Inclination test as specified in 4.7.33.

4.7.33 Inclined operation. The equipment shall be tested for inclined operation at 30 degrees from horizontal. The equipment shall be run at rated load for a period of not less than 1 hour at each of the following positions at an ambient temperature of 50°C.

(a) 30 degrees forward.
(b) 30 degrees backward.
(c) 30 degrees to left side.
(d) 30 degrees to the right side.

During the progress of this test it shall be ascertained that there is no apparent overheating of any drawers, modules or parts; dropping out of circuit breakers; or malfunction of operation of the equipment.

4.7.34 Grounding. The test shall be performed as follows: The equipment shall operate at nominal input voltage and at no-load and rated output load. Each input and output terminal shall be grounded in turn to ground as follows:

(a) Connect each terminal to ground through a 100,000 ohm resistance.
(b) Momentarily ground each terminal in turn five times by shorting one of the 100,000 ohm resistors in (a). Remove that short and then short the other 100,000 ohm resistor. Also ground each terminal for a period of 10 seconds and then remove the short.

4.7.35 Humidity. The equipment shall be tested in accordance with MIL-STD-810, procedure IV modified so that the maximum chamber temperature is 50°C.

4.7.36 Remote start-up and shutdown. The equipment shall be operated at nominal input voltage. The control of the equipment shall be from a remote control panel equivalent which is connected to the equipment with a cable having an equivalent resistance of a 500 feet type MSCU cable in accordance with MIL-C-915 and MIL-C-915/32. Equipment operation from the remote control panel shall be demonstrated.

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4.7.37 Stress level conformation. The electrical stresses of the parts in the equipment shall be tested to conform to the derated stress limits of 3.4.5.4 and NAVSEA 0967-LP-597-1011. Stress level shall be tested on representative parts.

4.7.38 Maintainability demonstration. The equipment shall be subjected to maintainability demonstration tests in accordance with MIL-STD-471, method 1, test B for mean corrective maintenance time (Mct) (15 samples). Maintenance tasks of the maintainability demonstration tests shall be performed by military or civil service personnel of the level and training available during operational use. After completion of each maintenance action, the equipment shall be operated for 15 minutes at rated-load. The time measured for each corrective maintenance task shall consist of the sum of the following steps: Fault location, fault correction, adjustment - if any and checkout, including the consulting of technical manuals and other allowed aids during this time.

4.7.39 Reliability. After completion of all other tests, a 2000 hour reliability test shall be accomplished in accordance with MIL-STD-2068. The test shall be performed with the unit at 50°C only. A vibration test is not required.

4.7.40 Portable tester test. The portable tester shall be connected to a 60 Hz, 115 V a.c. single-phase power source and connected to the frequency changer to demonstrate the checking of the logic circuits and the power switching drawer at the organizational level.

4.7.41 Endurance test. A continuous 200-hour endurance run shall be conducted on the equipment. The test conditions are as follows:

(a) Ambient temperature cycling - Operate equipment at 50 ± 5°C for 15 hours. Then reduce temperature to ambient of 25 ± 5°C for 5 hours. Repeat the cycle for 10 times continuously.
(b) Output load cycling - When the equipment is operating at 50 ± 5°C, the rated-load shall be applied continuously for 13 hours followed by 2 hours at no-load. When the equipment is operating at 25 ± 5°C, the rated load shall be applied continuously for 3 hours followed by 2 hours at no-load. The contractor may, for the purpose of energy management, schedule the rated-load operating hours and power down so as to avoid the normal peak utility load operating times.
(c) Output voltage and frequency - 450 V a.c. nominal, 400 Hz.
(d) Input voltage and frequency - 440 V a.c. nominal, 60 Hz.
(e) Data - The input current, input voltages, input frequency, output current, output voltage, output frequency and output power, test chamber ambient temperature, outside chamber ambient temperature and operating time, shall be recorded automatically every 15 minutes. Data shall be verified periodically (every 50 hours) by test personnel.
(f) If a failure should occur during the first 130 hours of operation, the cause of failure shall be recorded, corrected and the test continued. The equipment shall operate the final 70 hours without failure. If any failure occurs, the final 70 hours shall be repeated without failure.

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4.8 Inspection of packaging. Sample packages and packs, and the inspection of the preservation-packaging, packing and marking for shipment and storages shall be in accordance with the requirements of section 5 and the documents specified therein.

5. PACKAGING

(The preparation for delivery requirements specified herein apply only for direct Government acquisition. For the extent of applicability of the preparation for delivery requirements of referenced documents listed in section 2, see 6.6.)

5.1 Preservation-packaging, packing and marking. The equipment, technical manuals and data shall be preserved-packaged level A or C, packed level A, B or C as specified (see 6.2.1) and marked in accordance with MIL-E-17555.

5.1.1 Special marking. In addition to the marking requirements of 5.1, equipment tested and certified shall have the shipping containers marked in accordance with MIL-STD-740.

5.2 Tools. Tools shall be preserved-packaged, packed and marked in accordance with MIL-E-17555 or PPP-P-40 as applicable with the level specified for the intended use and destination as follows:

Destination	Levels	
	Preservation-packaging	Packing
On board (accompanying equipment)	A	C
Stock	A	B
Immediate use	C	C
INCO kits	A	C

5.3 Semiconductors or solid-state devices. Semiconductors such as diodes, transistors, integrated circuits as well as circuit boards or chassis in which they are incorporated, shall be individually packaged in a barrier bag in accordance with class E, style 1, type I or II, or class F, style 1, type I of MIL-B-117. MIL-B-117 bag material shall employ aluminum foil as a laminate of the bag barrier material. Bag closure shall be effected by heat sealing. Leads and terminals shall be protected from damage by means of the container (carrier) design, die cut inserts, or by the use of noncorrosive cushioning material. Leads and other projecting parts may be used for positioning, but shall not be subjected to loads or other stresses such as bending or twisting that can damage the entry seals. For level C preservation-packing, semiconductors or solid state devices subject to electromagnetic degradation shall be protected with a wrap of aluminum foil or a barrier bag employing aluminum foil as a laminate of the bag material. Leads and terminals shall be protected as specified herein.

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6. NOTES

6.1 Intended use. Solid state frequency changers covered by this specification are intended to convert Navy shipboard 60 Hz electrical power into 400 Hz electrical power for surface ships.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

(a) Title, number and date of this specification.
(b) Whether first article inspection is required (see 3.1 and 4.4).
(c) Preparation for shipment or storage required and additional marking if required (see 5.1).

6.2.2 Data requirements. When this specification is used in an acquisition which incorporates a DD Form 1423, Contract Data Requirements List (CDRL), the data requirements identified below shall be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved CDRL incorporated into the contract. When the provisions of FAR 52.227-7031 are invoked and the DD Form 1423 is not used, the data specified below shall be delivered by the contractor in accordance with the contract or purchase order requirements. Deliverable data required by this specification is cited in the following paragraphs.

Paragraph no.	Data requirement title	Applicable DID no.	Option
3.8	Drawings, engineering and associated lists	DI-E-7031	Level 2 Drawing number - contractor Design designation - contractor
4.1.1	Calibration system description	DI-R-7064	
4.4.1	First article inspection report	DI-T-4902	
4.5.1	Procedures, test	UDI-T-23732	
4.7.28 and 4.7.29	Summary, shop noise test results	UDI-T-23760	
4.7.31	Electromagnetic interference test report	DI-R-7062	

(Data item descriptions related to this specification, and identified in section 6 will be approved and listed as such in DoD 5000.19L., Vol. II, AMSDL. Copies of data item descriptions required by the contractors in connection with specific acquisition functions should be obtained from the Naval Publications and Forms Center or as directed by the contracting officer.)

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6.2.2.1 The data requirements of 6.2.2 and any task in sections 3, 4 or 5 of this specification required to be performed to meet a data requirement may be waived by the contracting/acquisition activity upon certification by the offeror that identical data were submitted by the offeror and accepted by the Government under a previous contract for identical item acquired to this specification. This does not apply to specific data which may be required for each contract regardless of whether an identical item has been supplied previously (for example, test reports ordered under 4.5.1).

6.3 First article inspection. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection as to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract.

6.4 Definitions.

6.4.1 Equipment. Equipment is defined as a solid state frequency changer having the power rating of 72 kW, 110 kW or 185 kW.

6.4.2 Unit. Unit is defined as each power rating of the solid state frequency changers; for example, a 110 kW frequency changer is one unit and a 185 kW frequency changer is another unit.

6.4.3 Cabinet. Cabinet is defined as a number of drawers physically joined together to perform a specific function.

6.4.4 Drawer. Drawer is defined as two or more parts or removable modules in one drawer. A drawer is a portion of a cabinet.

6.4.5 Module. Module is defined as two or more parts to form an entity as part of a drawer. Module can be replaceable as a whole, but having a part or parts which are individually replaceable.

6.4.6 Part. Part is defined as one piece, or two, or more pieces joined together which are not normally subject to disassembly, without destruction of intended use. Parts are the lowest level of breakdown.

6.5 Provisioning. Provisioning Technical Documentation (PTD), spare parts, and repair parts should be furnished as specified in the contract.

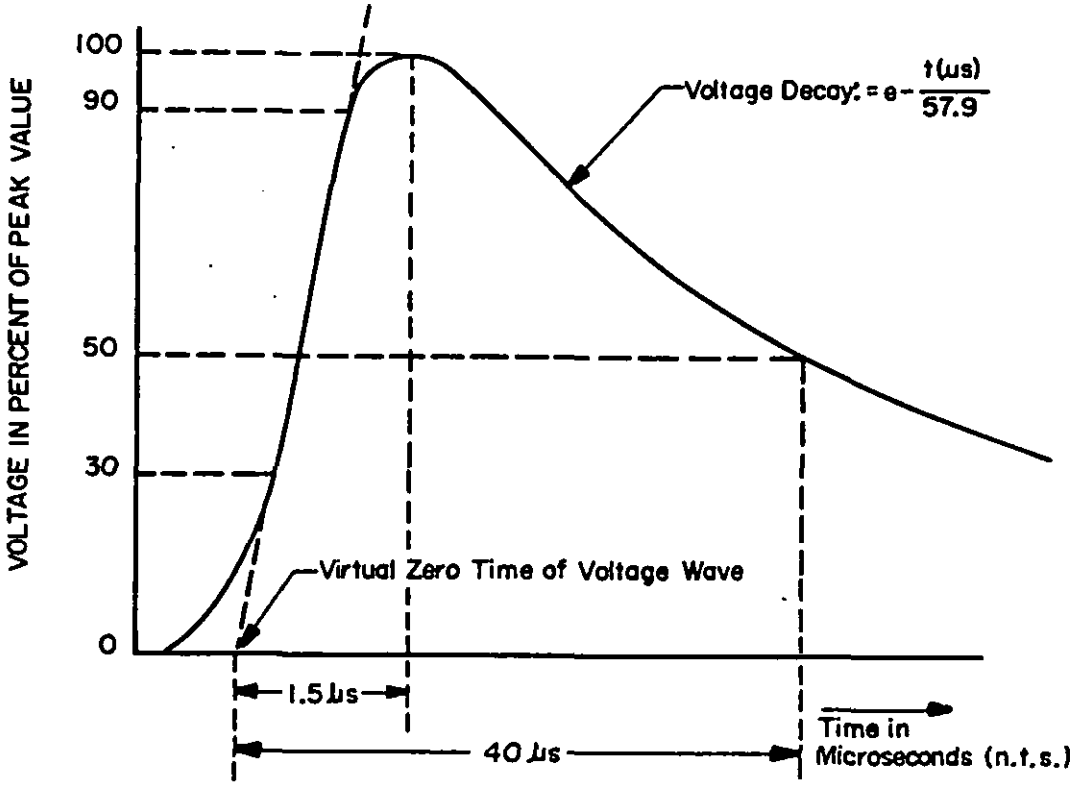
6.5.1 When ordering spare parts or repair parts for the equipment covered by this specification, the contract should state that such spare parts and repair parts should meet the same requirements and quality assurance provisions as the parts used in the manufacture of the equipment. Packaging for such parts should also be specified.

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6.6 Sub-contracted material and parts. The preparation for delivery requirements of referenced documents listed in section 2 do not apply when material and parts are acquired by the contractor for incorporation into the equipment and lose their separate identity when the equipment is shipped.

Preparing activity:
Navy - SH
(Project 6130-N243)

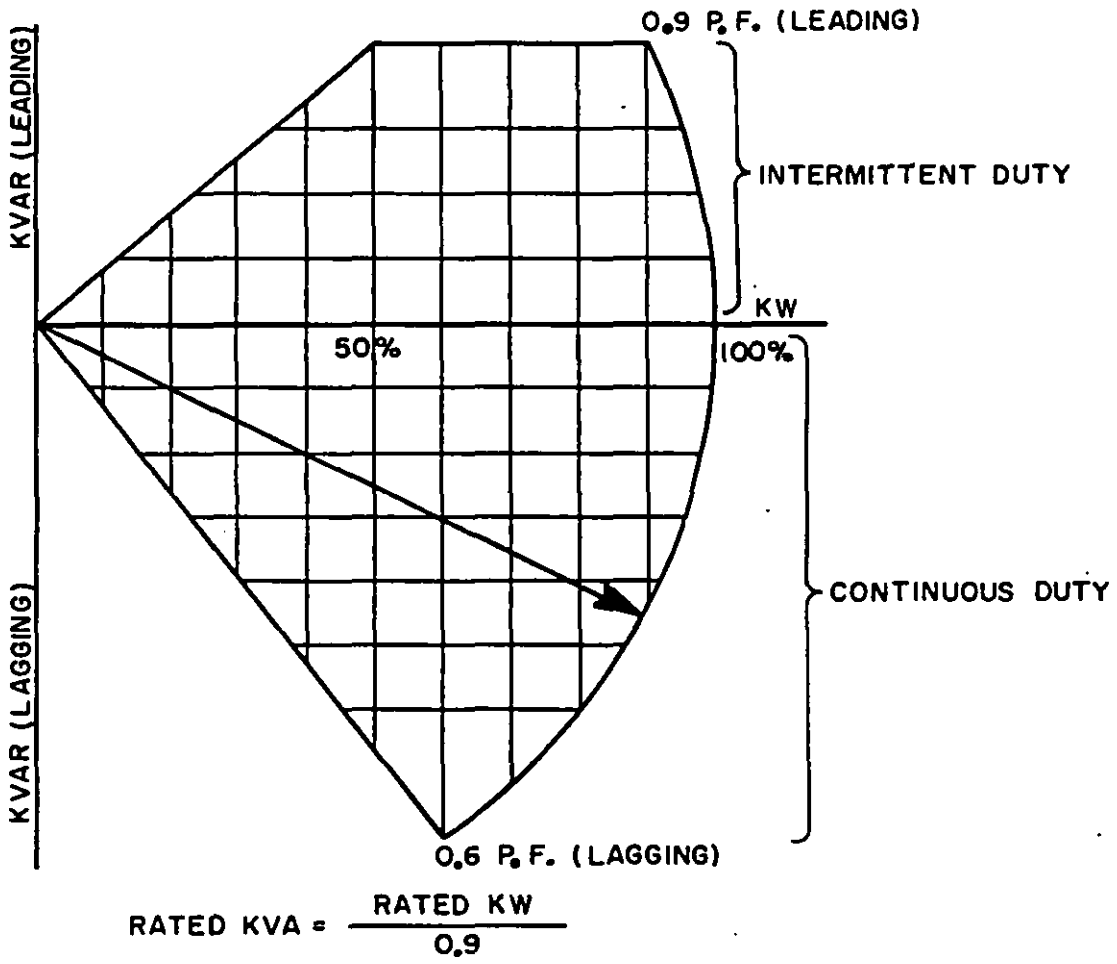
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SH 12424

FIGURE 1. Spike voltage (short time transient) wave shape.

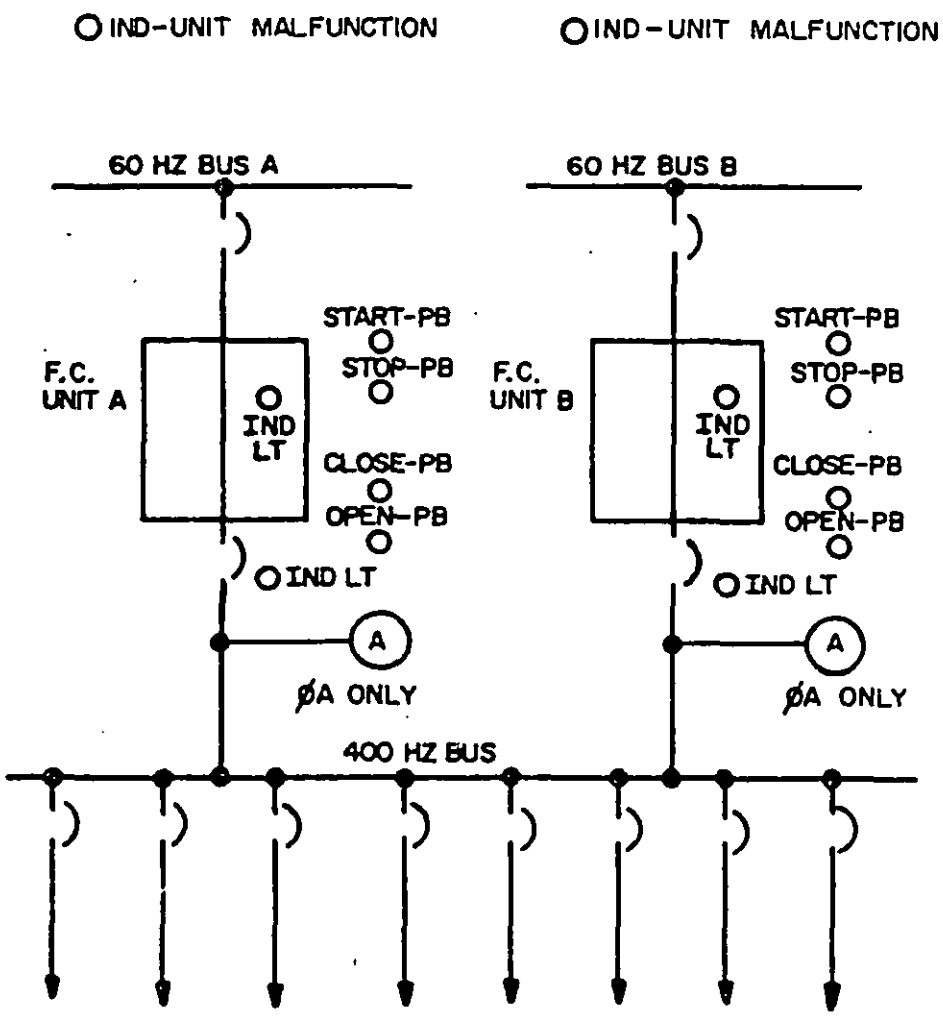
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SH 12426

FIGURE 2. Power rating.

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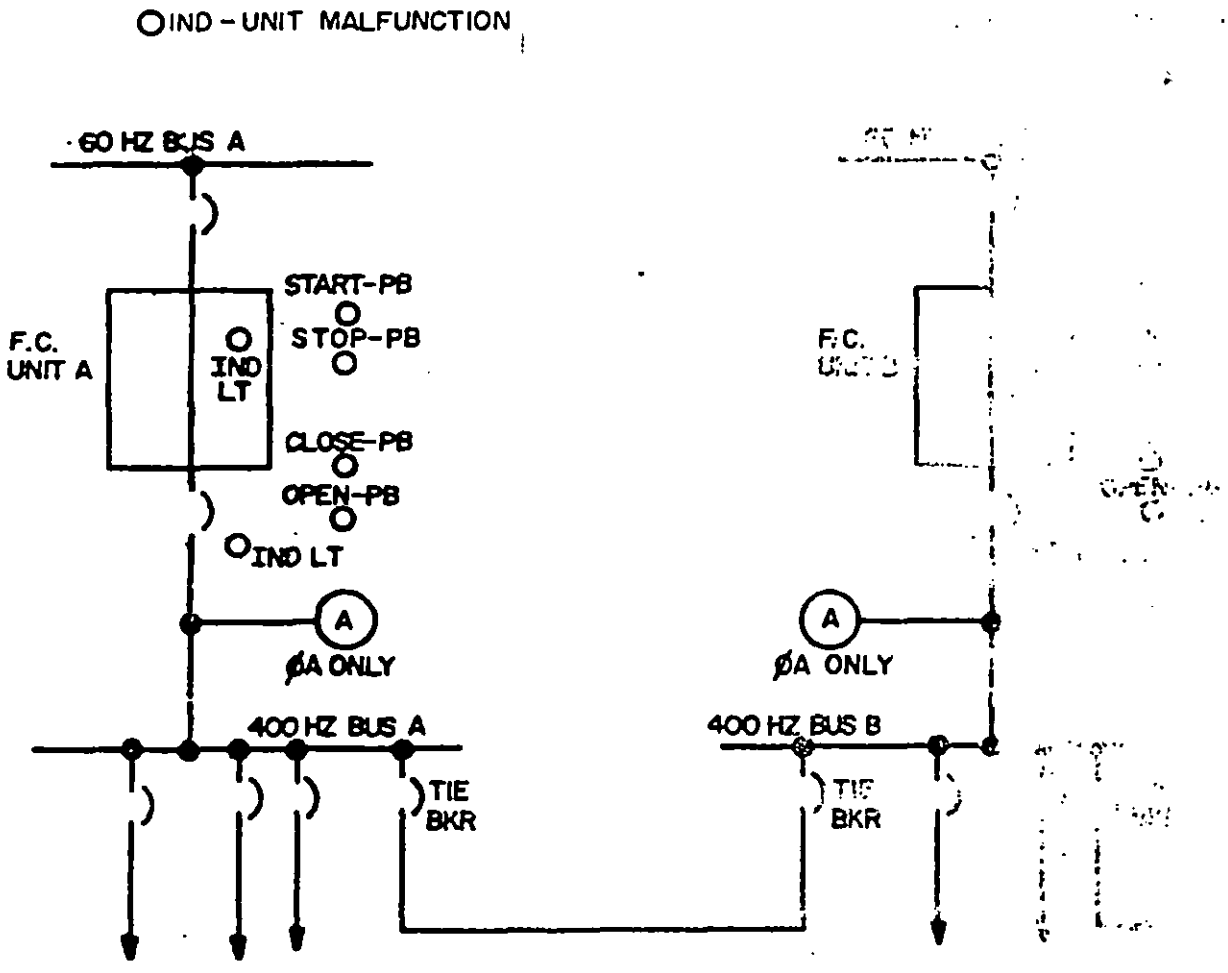


SH 11287

NOTE: Shown for two units only, but same arrangement applicable for up to six units.

FIGURE 3. Parallel operation - common bus.

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SH 11288

NOTE: Shown for two units only, but same arrangement applicable for up to six units.

FIGURE 4. Parallel operation - split bus.

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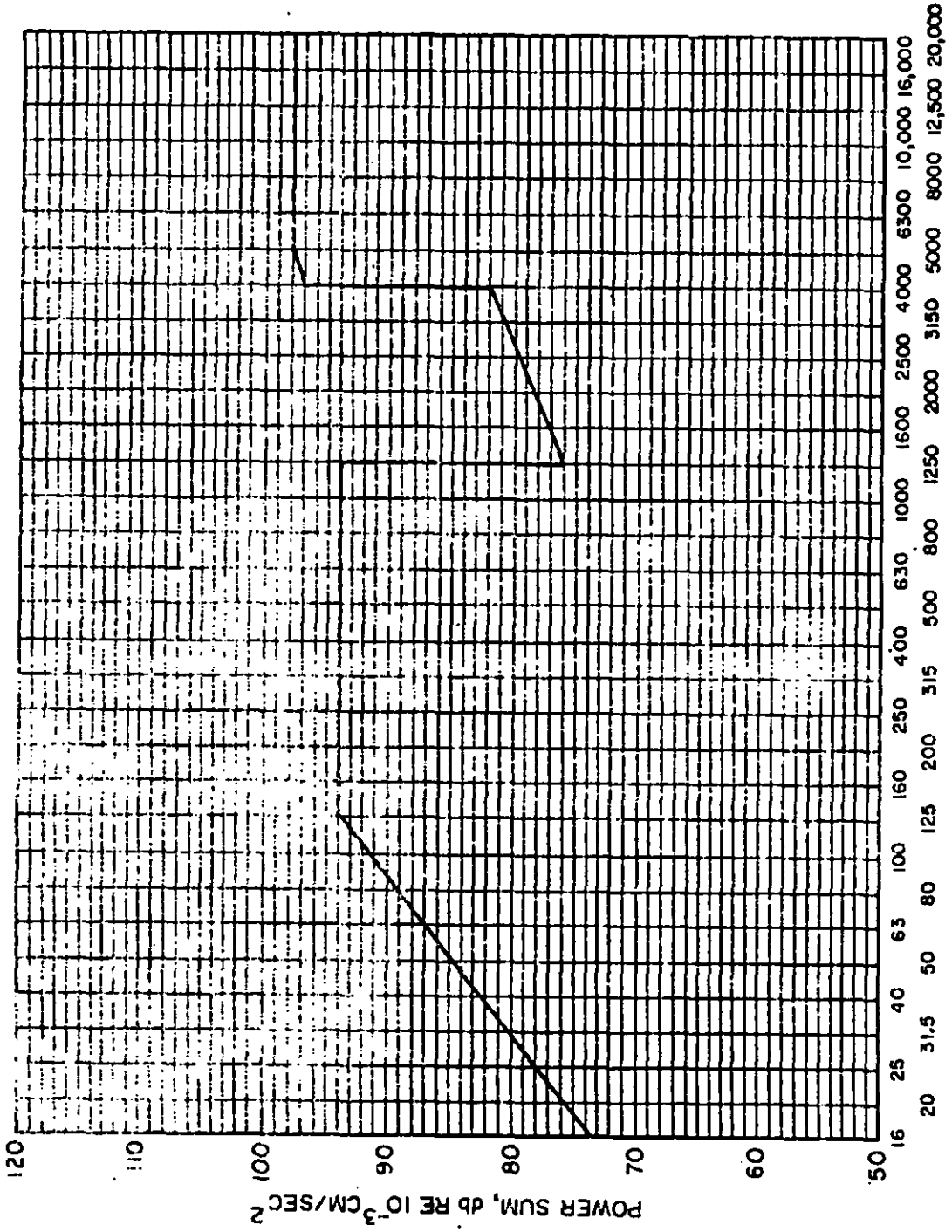


FIGURE 5. Structureborne noise plot.

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(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-F-24638(SH)	FREQUENCY CHANGER, SOLID STATE, AIR COOLED (NAVAL SHIPBOARD)

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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